

DESCRIPTION OF DESIGN DATA FOR ADHESIVES

Part: **1.4 – COMPOSITE MATERIALS**
Chapter: **1413**
Title: **DESCRIPTION OF DESIGN DATA FOR ADHESIVES**
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1 INTRODUCTION

This chapter lists and describes the physical and mechanical properties required for stress analyses with adhesive materials used to bond fibrous composites and cores.

The methods applicable calculate their design values from tests results data are presented in chapters 1480 "DERIVATION OF DESIGN DATA IN COMPOSITES."

Details on tests methods, on statistical methodologies, conventions and policies for presentation of data, are out of the scope of this document. See list of references for further information about these topics.

2 DEFINITIONS OF MATERIALS DESIGN DATA

2.1 IDENTIFICATION AND DESCRIPTION

Material identification and description shall be given in a similar manner as for other material types.

Short (informal) Name	
Provider and Commercial Name	
Material Designation(s) and Specification(s)	
Description	
Version	

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2.2 PHYSICAL PROPERTIES

Symbol	Units	Description
CPT	mm	Cured Ply Thickness, i.e. representative nominal thickness of the cured ply
ρ	kg/m ³	Density
T _{g-on}	°C	Glass transition temperature (onset) of the material in moisture saturated condition
MOL	°C	Maximum Operational Limit, or maximum continuous service temperature without material irreversible degradation

2.3 STIFFNESS AND STRENGTH OF ADHESIVE

Due to the low in-plane stiffness of the adhesive plies (compared with that of adherents) and the “free” expansion in out plane direction, thermal expansion coefficients are not normally necessary for stress analysis.

Stress-Strain curve in shear

The stress-strain curve of the adhesive in shear shall be obtained using test method ASTM D5656. For relevant ambient conditions, stress and strain values at the linear limit, at the “knee” point and at ultimate failure, shall be made available as approximation for the curve during stress sizing, in terms of mean and B-basis values.

Properties of adhesive in tension

Not frequently, the stress-strain curve of the adhesive material in tension could be available from tests data, using methods as the ASTM D638 or the D2095. Most frequently, in case properties in tension (Young modulus or ultimate tension strength) are needed, approximations derived from materials mechanics in isotropic materials and classical failure theories shall be used.

Tension Lap Shear strength

Adhesive qualification shall typically include results of mechanical tests of strength in single or double lap shear bonded joints with different types of adherents. These data shall typically be used to correlate and tune the material design values previously mentioned consistently with applicable stress methodologies in bonded joints.

Fracture toughness

Design Values for critical energy release rates, or fracture toughness, in fracture modes I (opening) and II (sliding / interlaminar shear) shall be provided for advanced numerical simulations.

Test methods used for obtaining these properties shall be typically AITM1-0053 for mode I and XXXXX for mode II.

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2.4 FATIGUE AND DAMAGE TOLERANCE (NO-GROWTH) OF BONDED JOINTS

Reserved for future use.

3 APPENDICES

OTHER ABBREVIATIONS AND SYMBOLS

Symbol	Units	Definition
1	-	Ply direction parallel to fibres in unidirectional tape materials, and in warp direction in fabrics, as subscript
2	-	Ply direction normal to fibres in unidirectional tape materials, and in weft direction in fabrics, as subscript
3	-	Ply through-thickness direction, as subscript
EKDF / K_{EKDF}	-	Environmental Knock Down Factor to account for degradation of a mechanical property with material saturated in an ambient (temperature and humidity) different to the reference conditions of 20°C (Room Temperature or RT) and dry material
K_B	-	B-basis Knock Down Factor for the minoring to the mean value of a mechanical property in order to get a safe design allowable value accounting for natural dispersion
TBC	-	To Be Confirmed

REFERENCES

	Document Reference	Issue	Title
Ref. 1	[SAE] CMH-17	2012	COMPOSITE MATERIALS HANDBOOK
Ref. 2	[Airbus] PP0300181	Is. 2	Composite statistical method to determine B-basis value from small data sets"
Ref. 3	[Airbus] RP0412191	Is. 1	Normalization rules for composite properties

SOFTWARE

Program	Version	Description

Table 1. Software programs referenced in this chapter

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RECORD OF REVISIONS

Revision Date	Authors	Change Reason	Pages/Sub-chapters affected
0.9 24/07/2020	G. Baños	First issue	All pages

Table 2. Record of Revisions: authors, change reasons and sections/pages affected

APPROVAL SIGNATURES TABLE

Dpt. \ Site	Getafe	Manching
Stress Methods (TEPAP-TL4)	G. Baños 24/07/2020	Florian Glock XX/07/2020

Table 3. Approval signatures table for last revision of this chapter